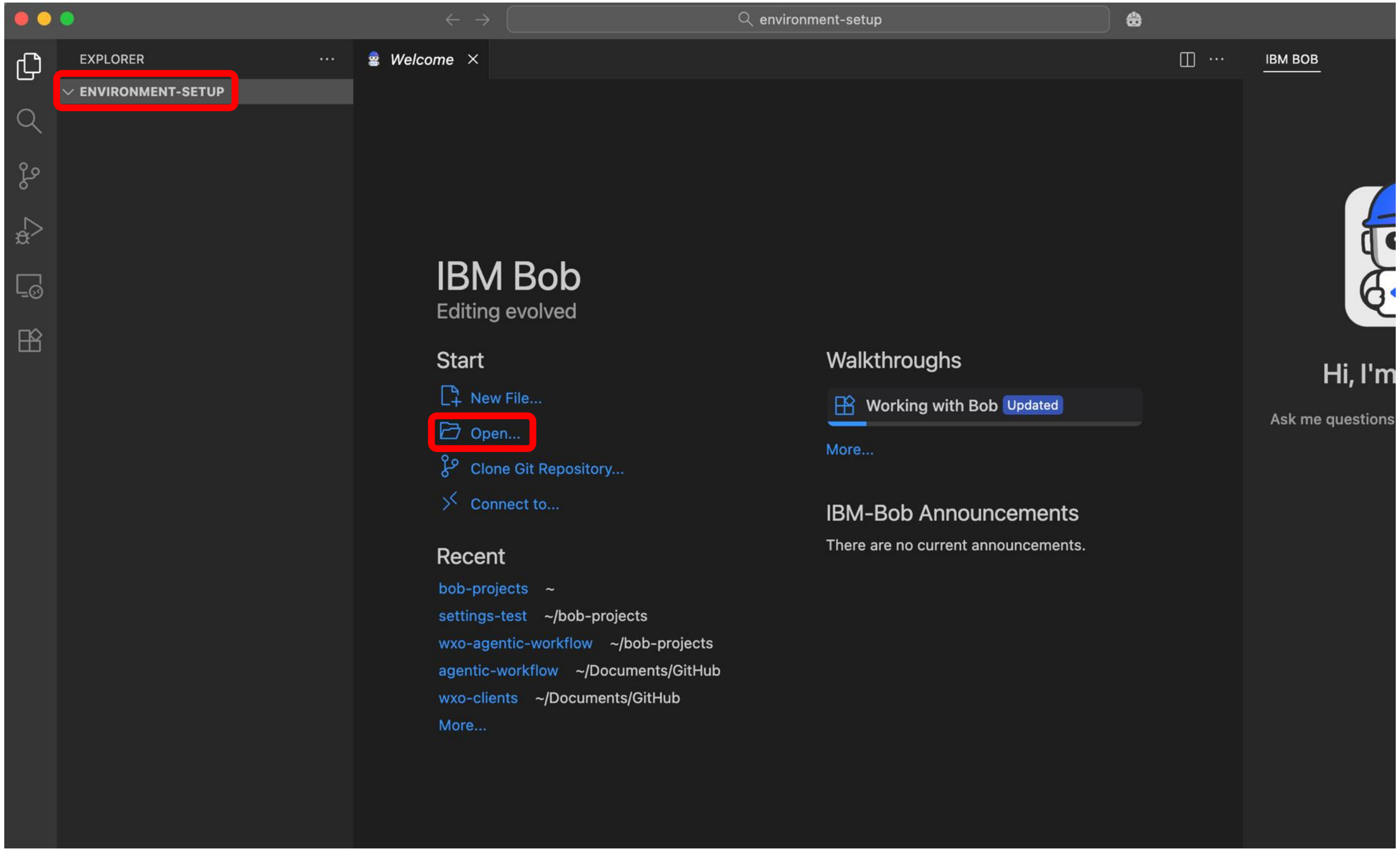
Environment setup instructions

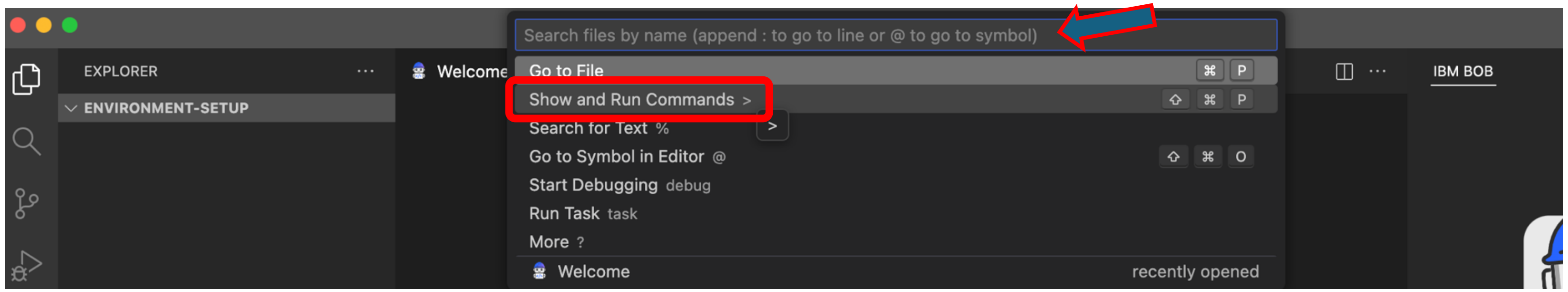
Getting your wxO ADK installed (1)



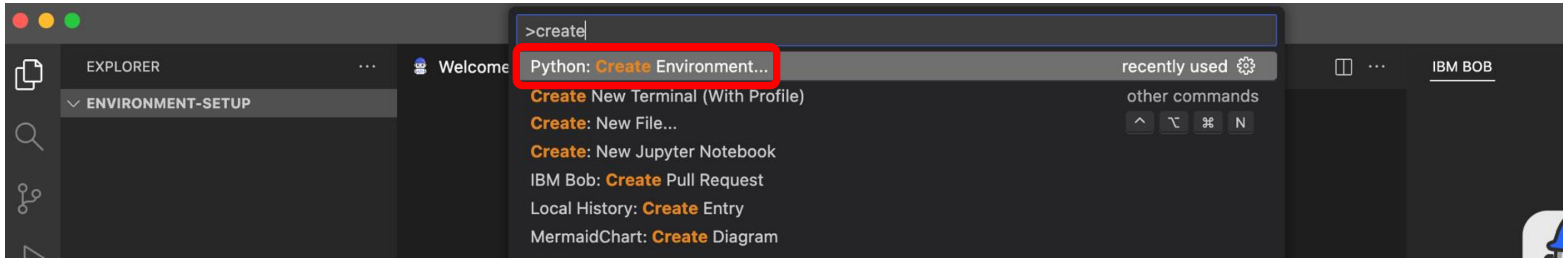
- Open New Window within Bob IDE (File → New Window ⇧⌘N) to start from a clean workspace
- **Create** and **Open** a new folder for your project (name it properly, here we use the folder named *environment-setup* just as an example)

NOTE! When preparing for the first hands-on lab in Paris T3, name the folder as "**wxo-agentic-workflow**"

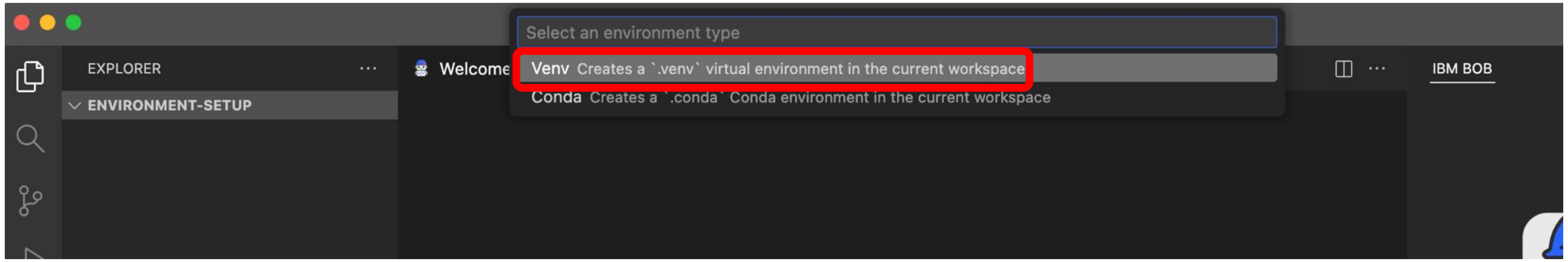
Getting your wxO ADK installed (2)



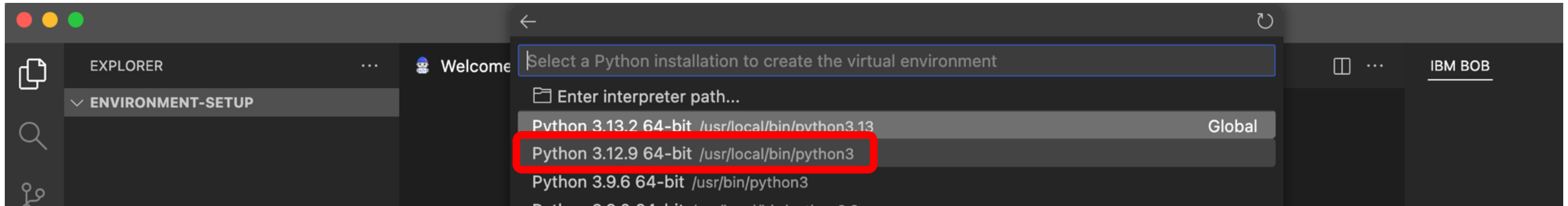
- Click the search bar on the top and select ***Show and Run Commands***



- Type in "***create***" and select **Python: Create Environment...**

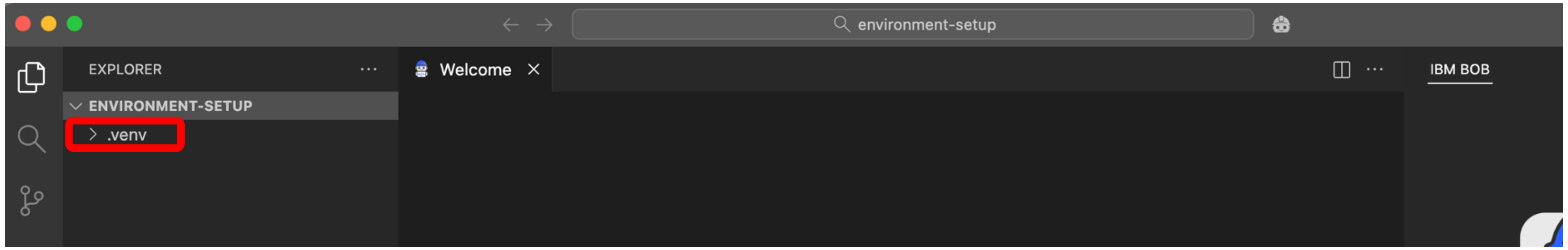


- Select **Venv** if given some options (depends on what you have installed to your computer)

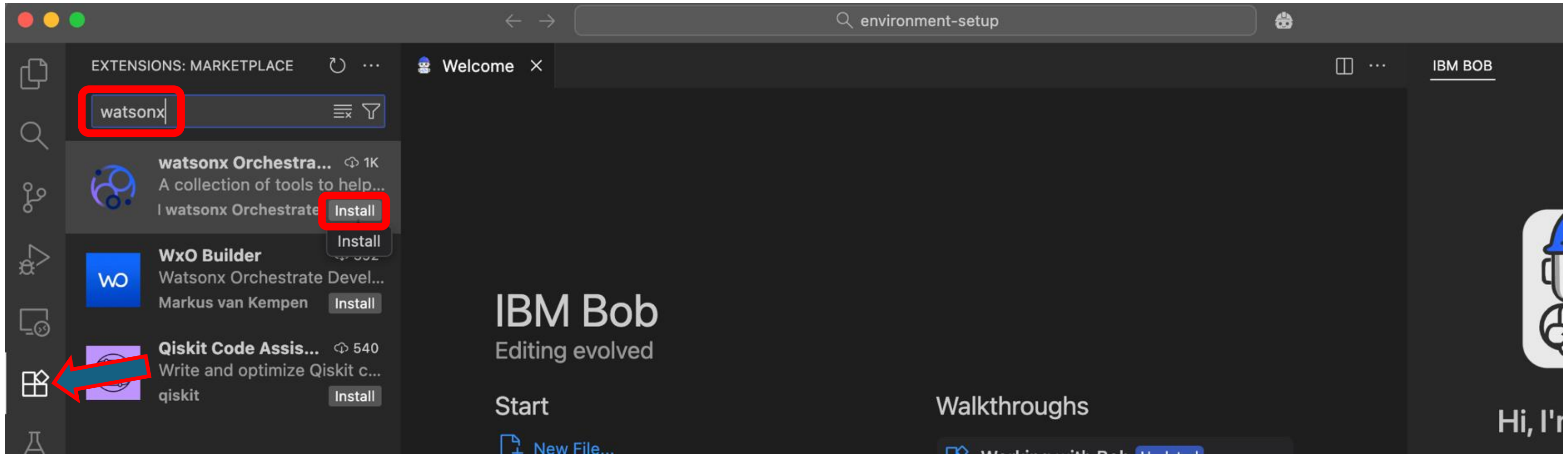


- Select your Python installation, 3.11.x – 3.13.x (**3.12.9** recommended)

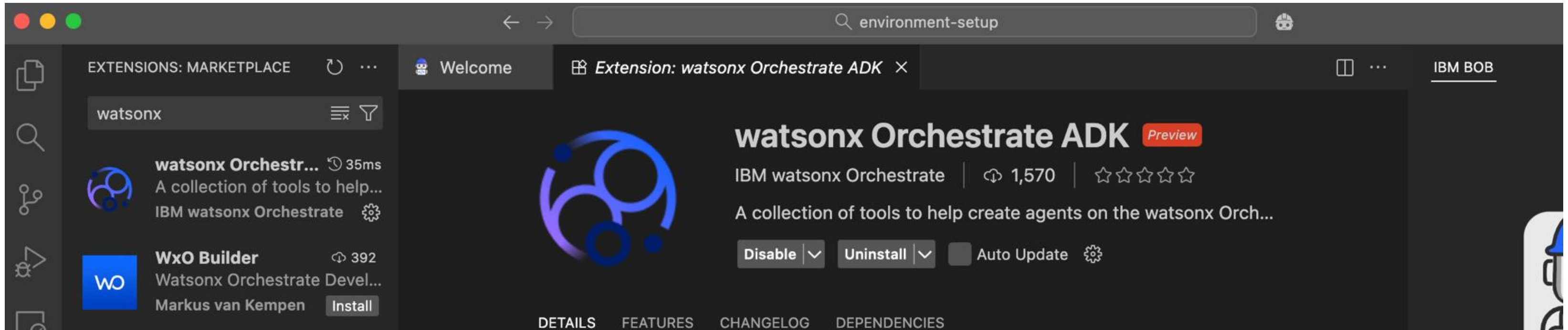
Getting your wxO ADK installed (2)



- The Python virtual environment will be installed and you'll see the **.venv** folder under your project folder in a couple of seconds

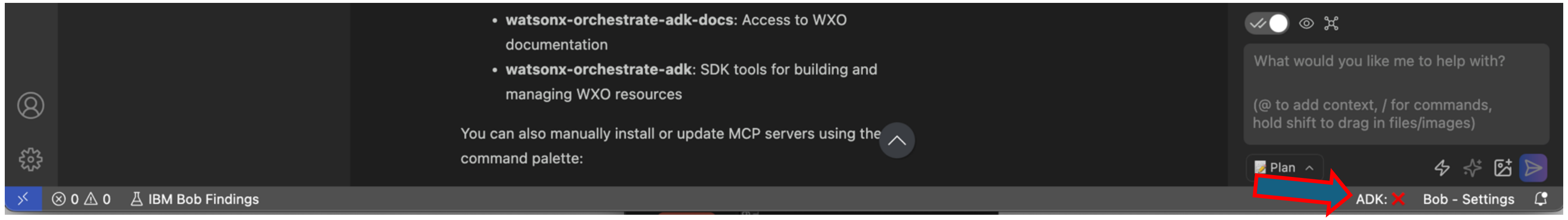


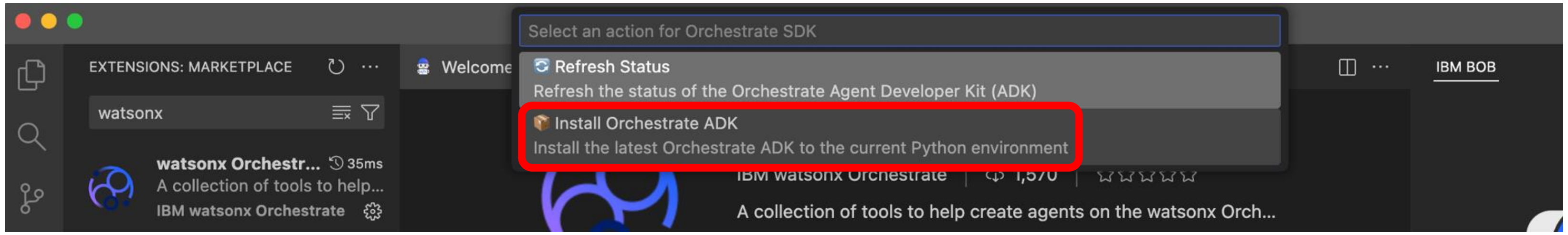
- Click the **Extensions** icon from the menu bar on the left, search for "***watsonx***" and click **Install** on the watsonx Orchestrate ADK extension

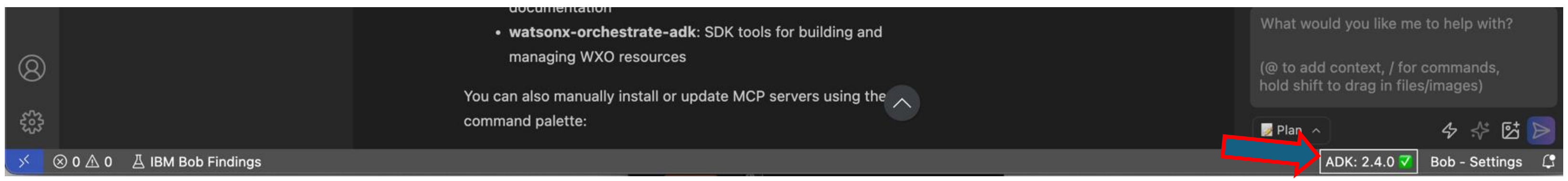


- This will install the extension (note that the extension is still in ***preview***)

Getting your wxO ADK installed (3)







- Now that you have the wxO ADK extension installed, you can see the ADK information on the bottom right of Bob IDE window
- Click the **ADK: X** –area and this will open the search bar with two options, select "**Install Orchestrate ADK**"
- After a while you will see the latest ADK version installed to your active Python virtual environment

Let's configure the wxO MCP servers to Bob!

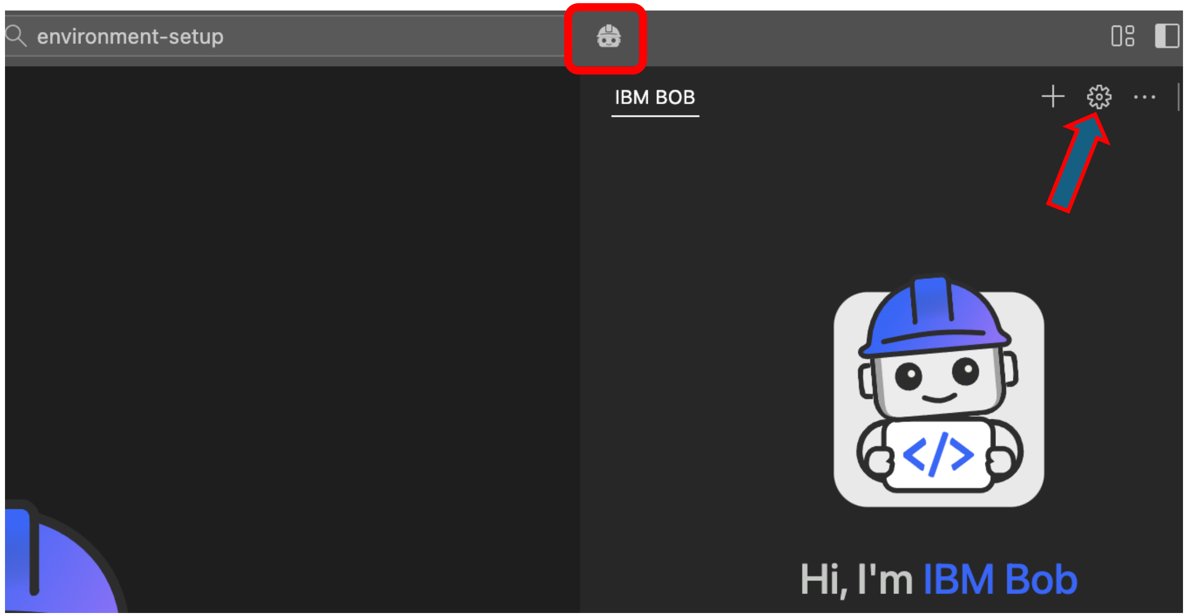
Configuring the wxO MCP servers for Bob (1)

- IBM watsonx Orchestrate (wxO) provides two different MCP servers that we can add to IBM Bob's "knowledge": 1) the documentation server (wxo-docs) that Bob can access and learn how create agents, tools, connections, etc. with wxO and 2) wxo-adk to operate the active wxO environment – import/export agents and tools, run configuration changes, etc.

- These MCP servers are already available in IBM Bob Marketplace, but currently (before GA) you will need an IBM network connection to access it – in Paris we should have it, and you can use the following instructions (if having troubles, check slide 13 for manual setup)

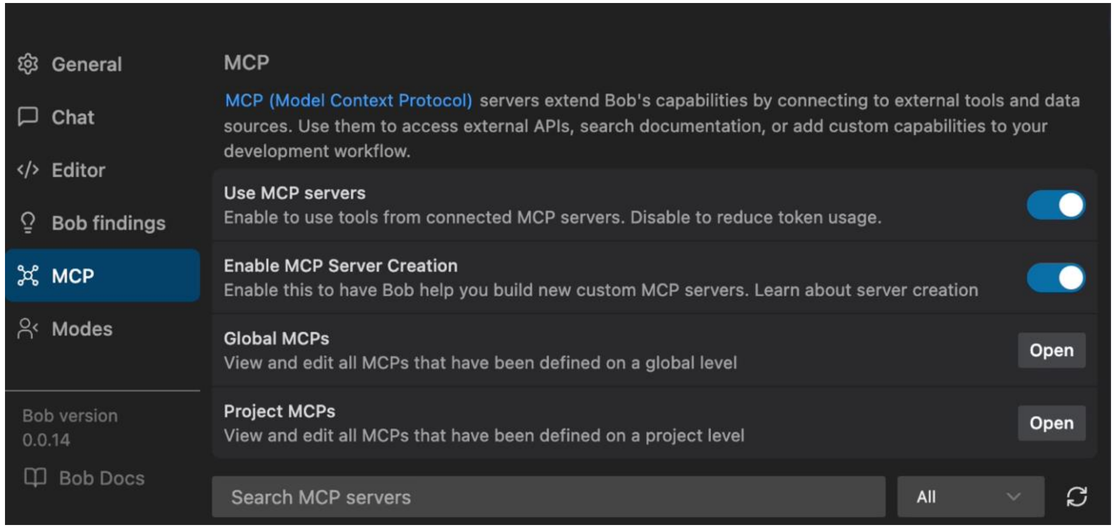
- Open the IBM BOB side panel if not already open by clicking the Bob icon besides the search bar

- Then click the settings icon to access all the IBM Bob IDE settings

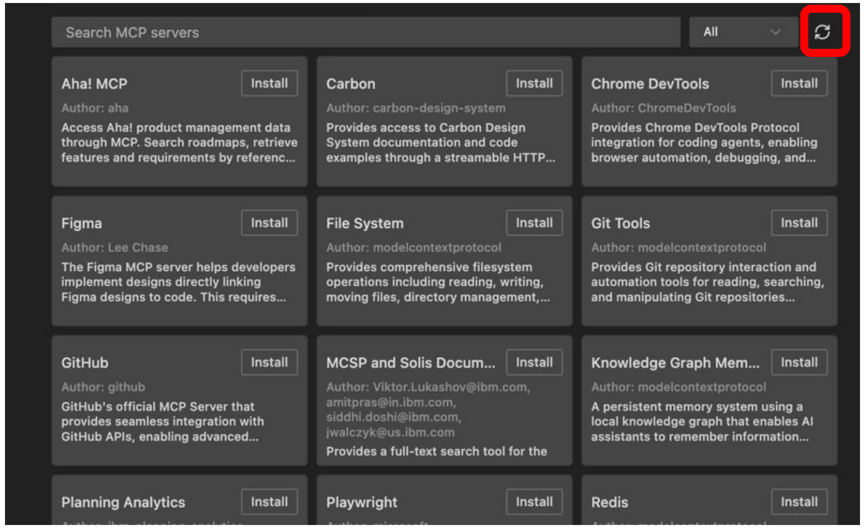


Configuring the wxO MCP servers for Bob (2)

- Select the **MCP** category and make sure you have both *Use MCP Servers* and *Enable MCP Server Creation* toggles enabled



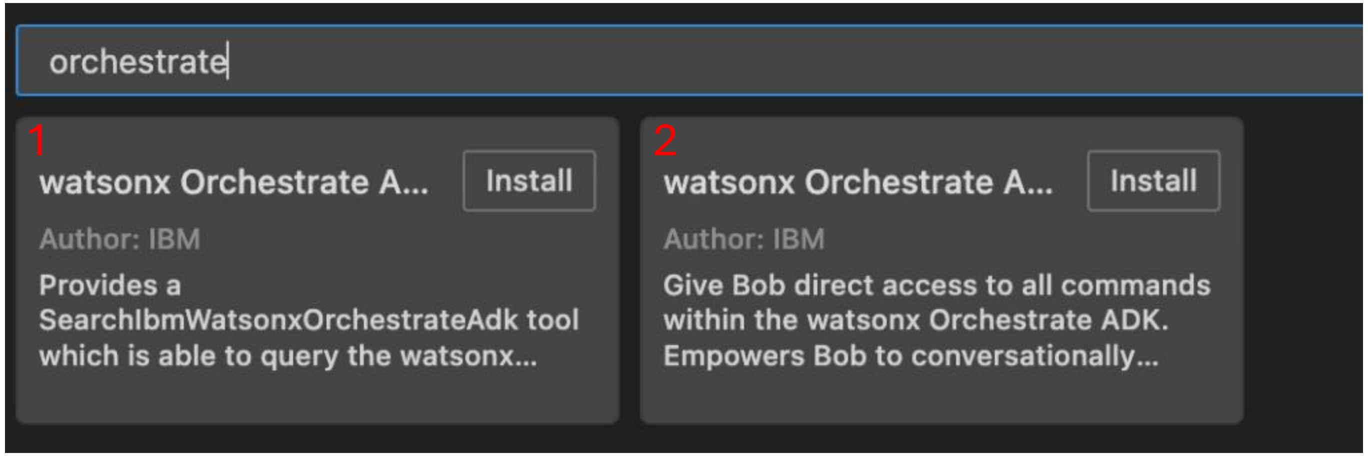
- You should see the currently available MCP servers under the MCP settings, if not, you can click on the refresh icon



- **Note**: if you cannot see the MCP servers, you might not have the IBM network connection. If the problem persists, go to slide X add the servers manually.

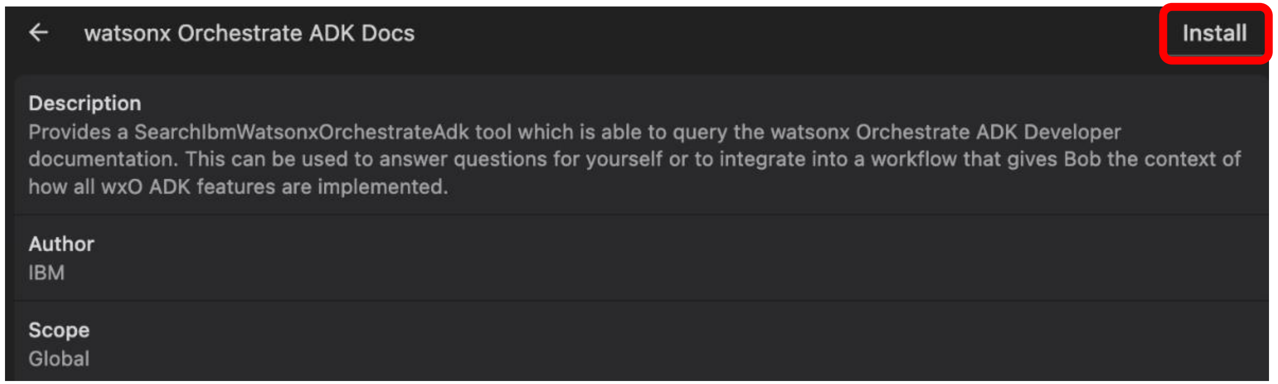
Configuring the wxO MCP servers for Bob (3)

- Search for "*orchestrate*" and you will find the 2 available MCP servers for wxO



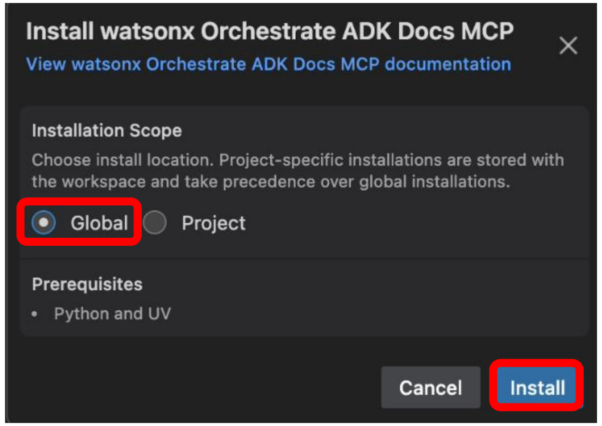
1. *The first one provides a search tool for Bob to search all the available documentation on wxO Agent Development Toolkit (ADK*
2. *The second one provides Bob with direct access to all the commands within the ADK, allowing Bob to interact with your development environment*

- Click the first tile (with the documentation search tool) to see details for the MCP server and then click **Install**

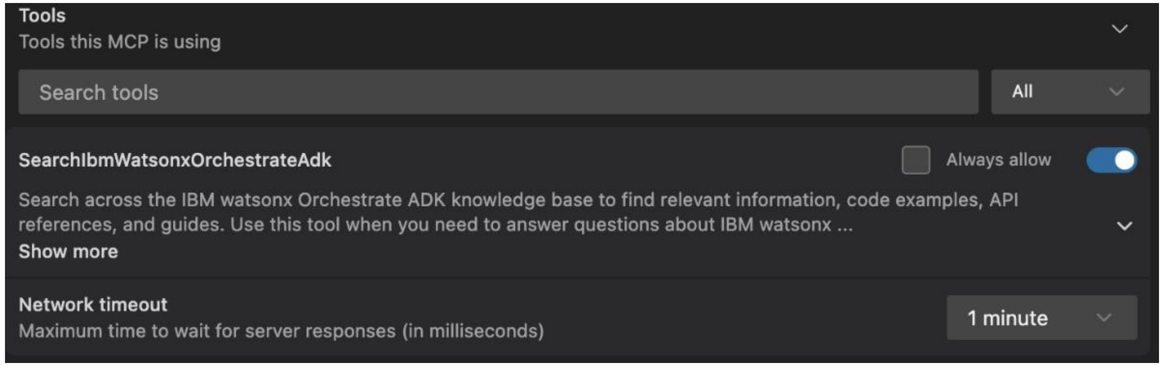


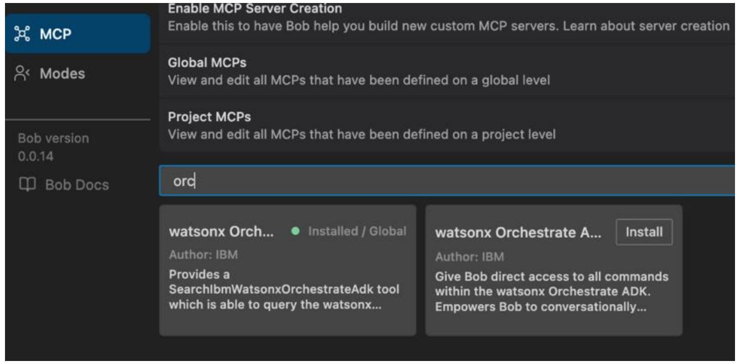
Configuring the wxO MCP servers for Bob (4)

- Select **Global** as the installation scope and click **Install**



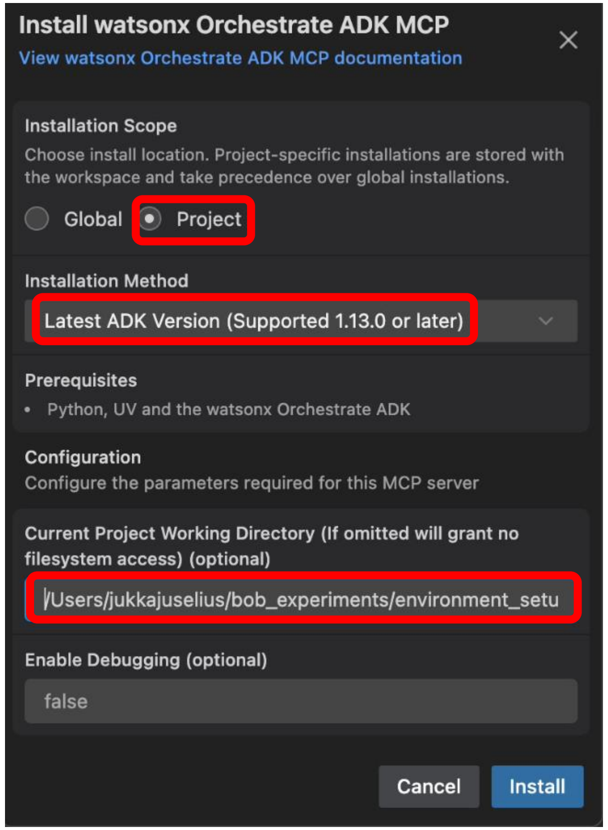
- Leave the settings for the MCP server as they are and navigate back to base MCP servers view and you can now see the wxO documentation server installed



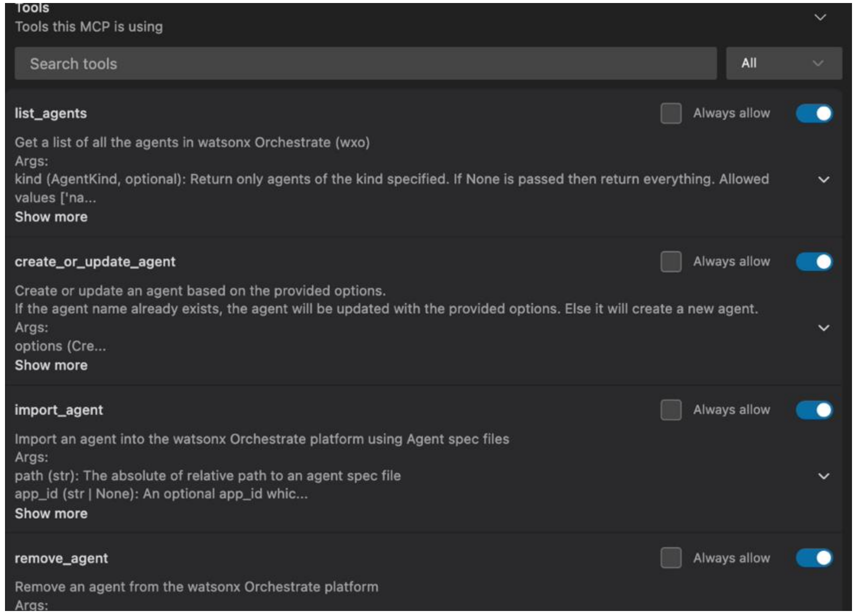


Configuring the wxO MCP servers for Bob (5)

- Next, install the other wxO MCP server as well – keep the scope as **Project**, select **Latest ADK Version** for the Installation Method and provide **the path for your workspace folder**, click **Install**

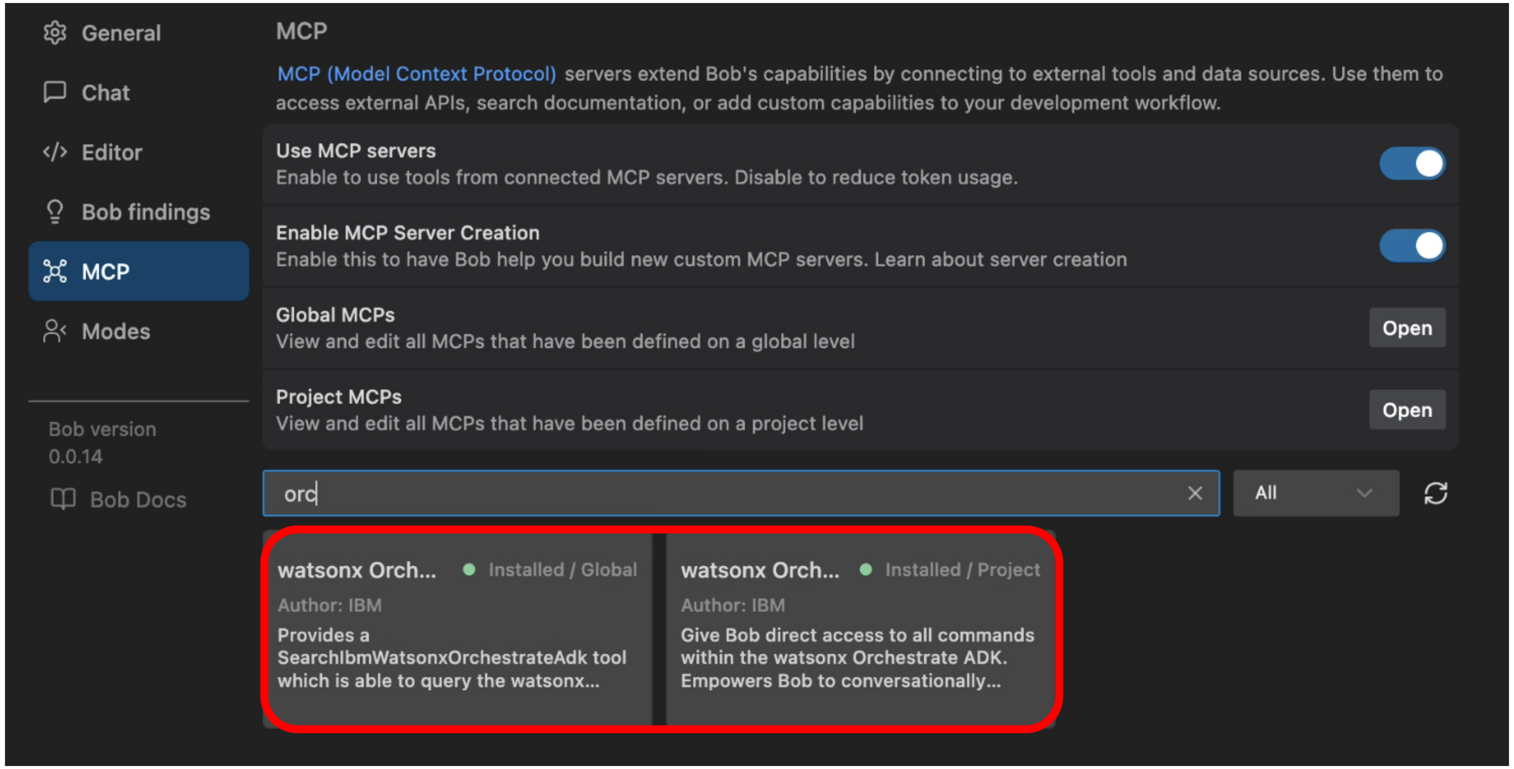


- Keep the default configuration for enabled tools as it is: all enabled, none always allowed



Configuring the wxO MCP servers for Bob (6)

- You should now have both servers connected



- *One more thing to do*: add a new wxO Agent Architect mode to Bob, so that it can make use of the MCP servers

Installing wxO MCP servers manually (1)

- Step 7 from this tutorial: https://developer.ibm.com/tutorials/build-agents-mcp-tools-watsonx-orchestrate-using-bob/

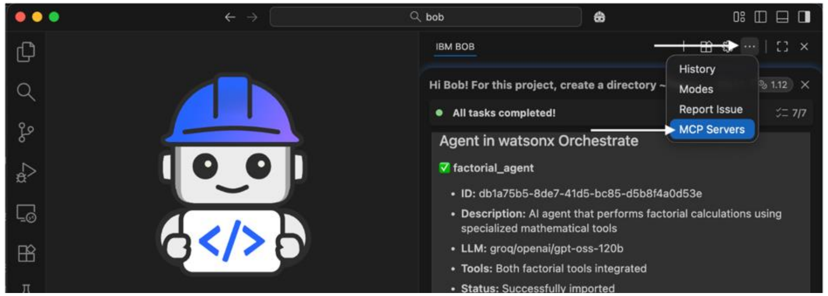
(Optional) Step 7. Connect Bob to watsonx Orchestrate MCP servers

After completing the agent workflow using Bob, you can set up Bob to connect to watsonx Orchestrate MCP servers. Connecting Bob to these servers makes it easier to work with agents and tools in watsonx Orchestrate. These servers provide resources for building and managing agents:

- **wxo-docs**: Provides access to public documentation for the watsonx Orchestrate ADK.
- **orchestrate-adk**: Provides access to the watsonx Orchestrate software development kit (SDK) for creating agents and tools.

To set up Bob to access watsonx Orchestrate MCP servers, complete the following steps:

1. In Bob, click **Open folder** from the File menu and then select the current working directory that Bob created and click **Open**.
2. In the right pane, click **...** > **History** and then select the last conversation with Bob for **Workspace: All**.
3. In the right pane, click **...** > **MCP Servers**.



5. Add the following configuration to configure the connection to the watsonx Orchestrate MCP server that has access to the documentation and then save the file.

```
{
 "mcpServers": {
     "wxo-docs": {
     "command": "uvx",
     "args": [
         "mcp-proxy",
         "--transport",
         "streamablehttp",
         "https://developer.watson-orchestrate.ibm.com/mcp"
     ],
     "alwaysAllow": [
         "SearchIbmWatsonxOrchestrateAdk"
     ],
     "disabled": false
     }
 }
 }
```

6. In the right pane, click **Edit Project MCP** and add the following configuration to enable access to the watsonx Orchestrate MCP server for managing agents and tools.

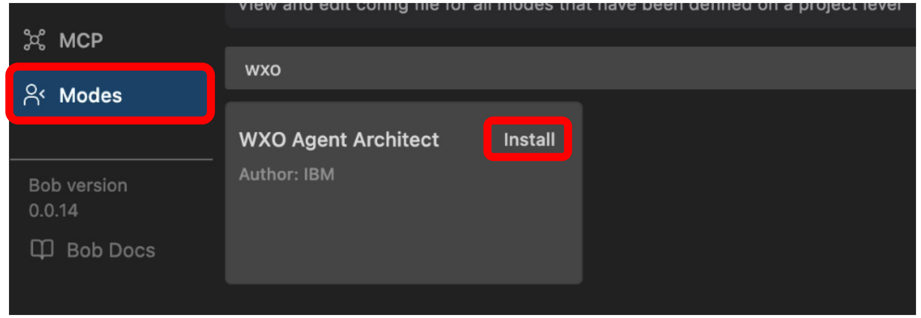
Note: Replace **WXO_MCP_WORKING_DIRECTORY** with your project path (for example: **~/Documents/bob/mcp-wxo**) and then save the file.

```
{
   "mcpServers": {
     "orchestrate-adk": {
       "command": "uvx",
       "args": [
         "--with",
         "ibm-watsonx-orchestrate==2.1.0",
         "ibm-watsonx-orchestrate-mcp-server==2.1.0"
       ],
```

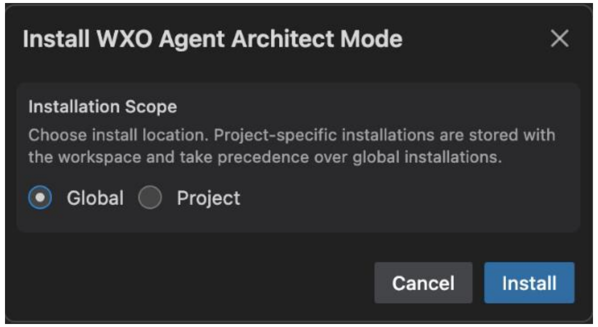
Configuring the wxO Mode for Bob (1)

- Now that you have the wxO MCP servers available we can also add a new mode for Bob, so that we can select when we want Bob to use the servers

- Switch to **Modes** view in the Bob settings, search for wxo and click **Install** on the WXO Agent Architect



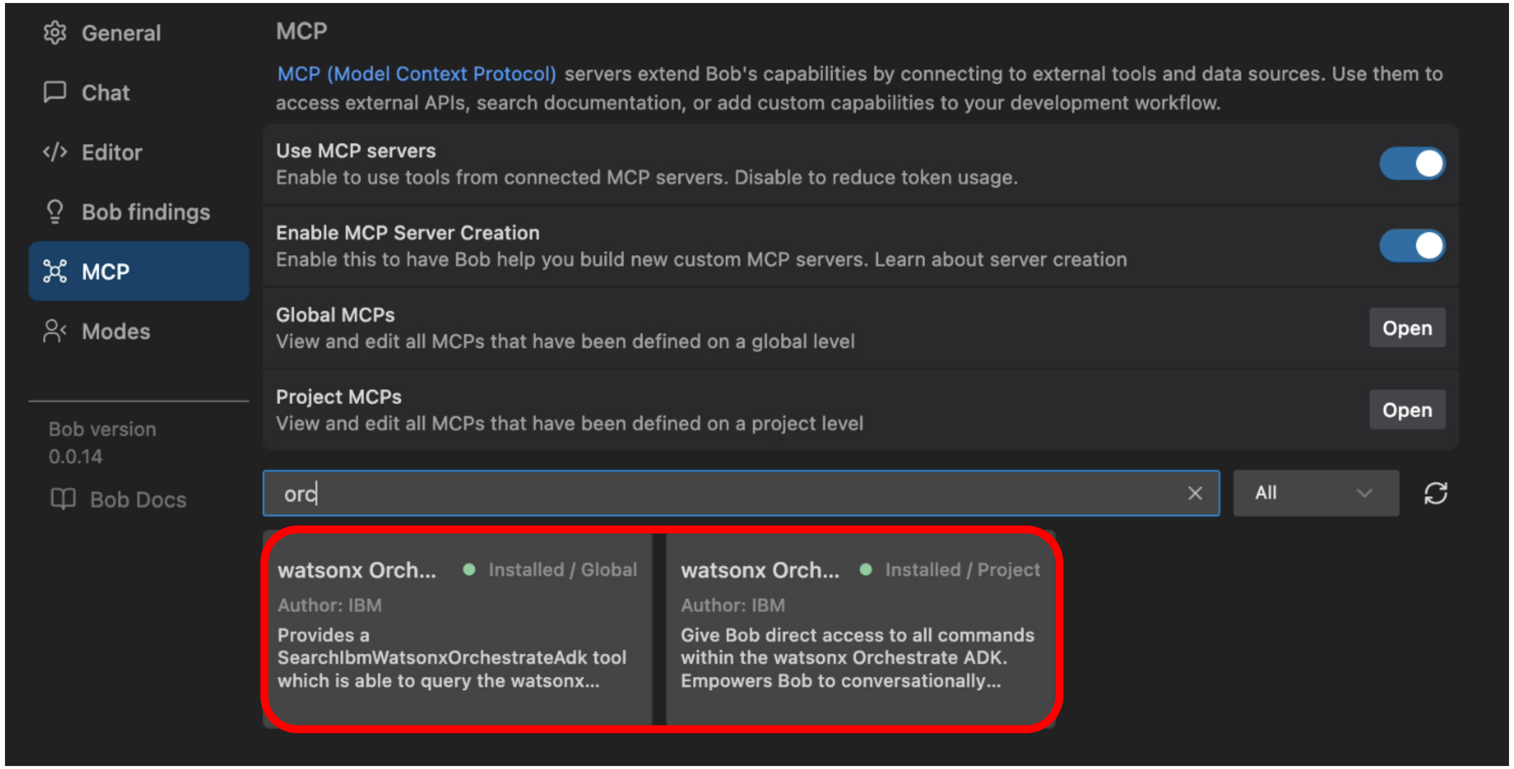
- Select **Global** as the Installation Scope and click **Install**



Next, Let's connect your ADK to the wxO environment you want to use!

Configuring the wxO MCP servers for Bob (6)

- You should now have both servers connected



Let's connect your ADK to the wxO environment you want to use!

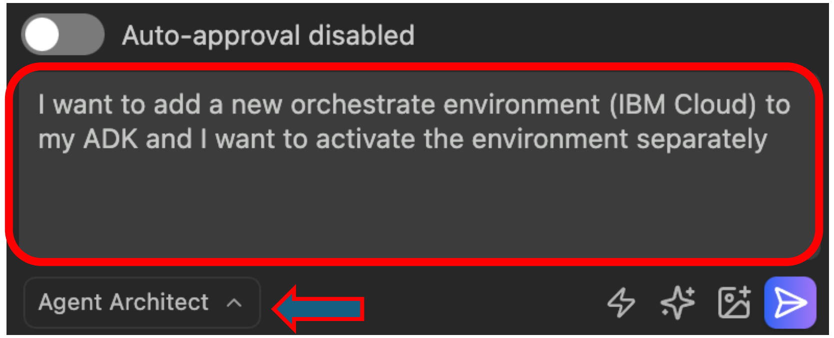
Connecting ADK to your wxO environment

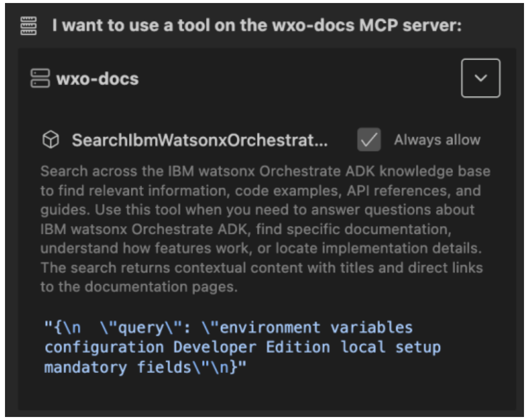
2 options:

1. Install wxO runtime – Developer Edition – locally (min. 16 GB mem needed, 32 GB recommended)
2. Use existing wxO environment for example from IBM Cloud (SaaS)

Option 2: Connect to wxO SaaS (1)

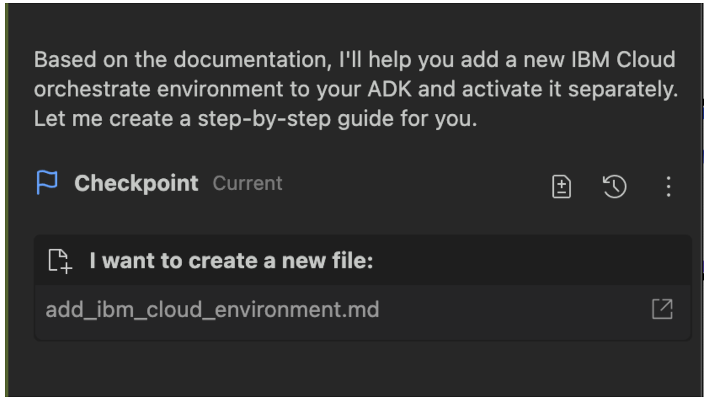
- Instead of running wxO server locally (wxO Developer Edition) you can use existing wxO environment and connect your ADK to it
- If you already have watsonx Orchestrate MCP servers defined for Bob, you can ask Bob to help you define the environment connection 😄
  - In Bob chat, ask Bob to create the env file (**Note**: make sure to have the mode selected that allows Bob to use the wxO MCP servers): "***I want to add a new orchestrate environment (IBM Cloud) to my ADK and I want to activate the environment separately***"
  - Bob should ask access to wxo-docs MCP server to use the watsonx Orchestrate documentation, **Approve** access when asked and soon you will see Bob starting to create instructions for you
  - Bob might ask if you already have the instance URL and the API key, select the answer that you still need to get them from IBM Cloud



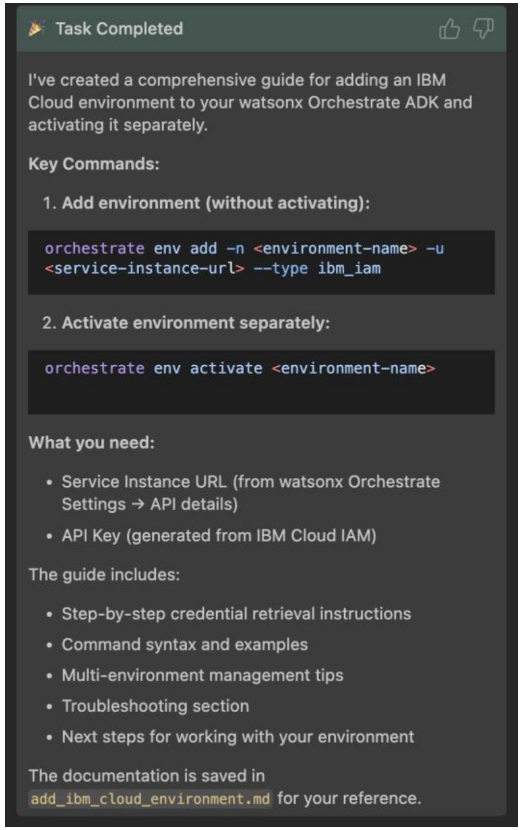


Option 2: Connect to wxO SaaS (2)

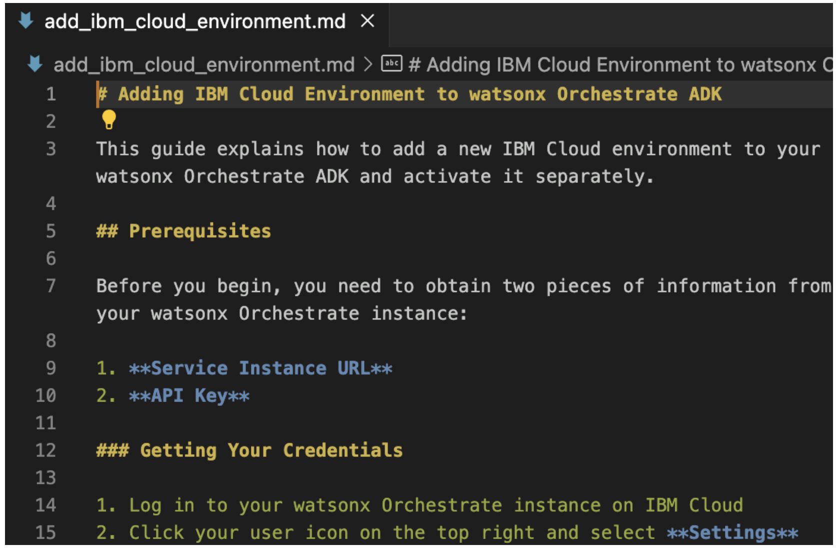
- Bob will the ask permission to create/write a new file, give the permission (**Save**)



- Bob will then summarize the task for you (task completed)



- Bob will also open the .md file it created for you – these are the instructions you can follow the add the wxO environment from IBM Cloud

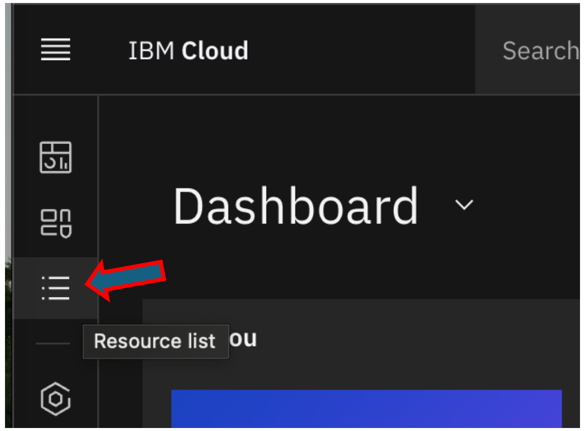


Option 2: Connect to wxO SaaS (3)

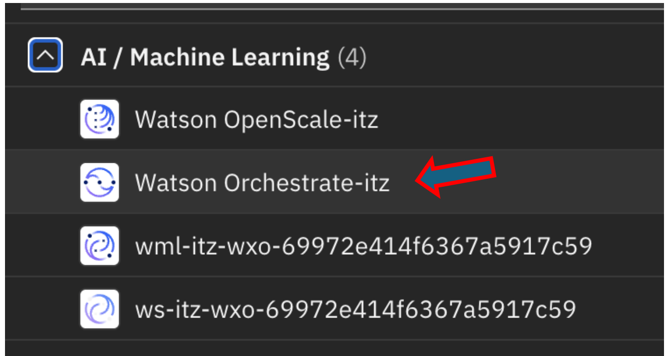
- Now you need to find out your wxO instance URL and API key for your IBM Cloud wxO instance as described in the generated instructions

- **Login** to IBM Cloud account that you should have been invited (when you provisioned your IBM Cloud wxO environment or when your instructors added you to one of the pre-provisioned accounts)

- Select **Resources list** from the menu bar on the left

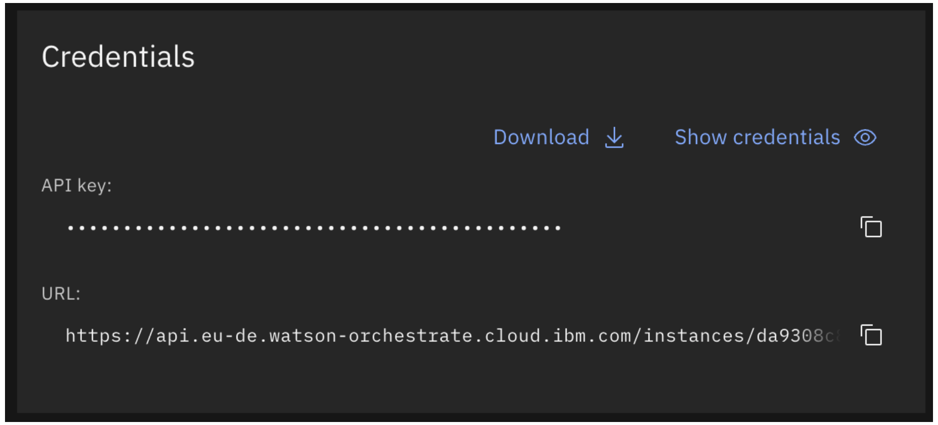


- Open the collapsible menu for **AI / Machine Learning** and then click on the **Watson Orchestrate-itz** service to open it



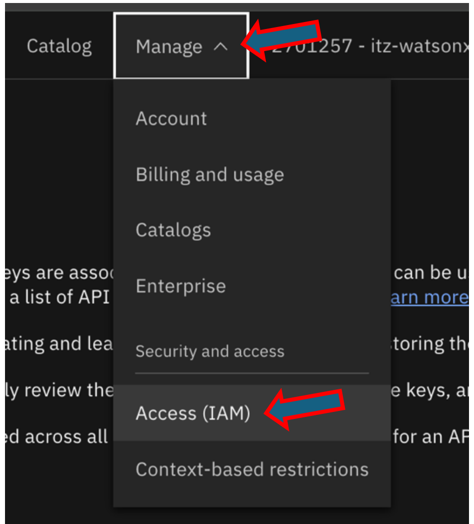
Option 2: Connect to wxO SaaS (4)

- Copy the service **URL** and store it so that you can easily access it in the next steps



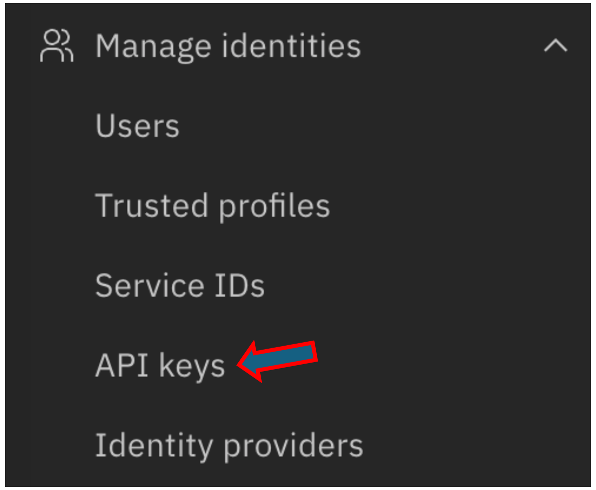
- Unfortunately, it seems that the default API key is not working for this ITZ environment as it should, and you need go and create your own API key to use

- Open the **Manage** menu from your IBM Cloud page and select **Access (IAM)**

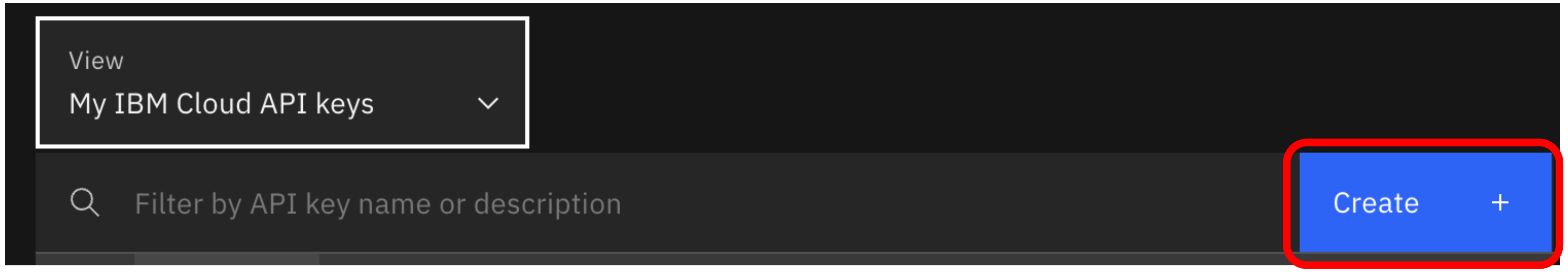


Option 2: Connect to wxO SaaS (5)

- Select **API keys** from the men on the left (under Manage identities section)

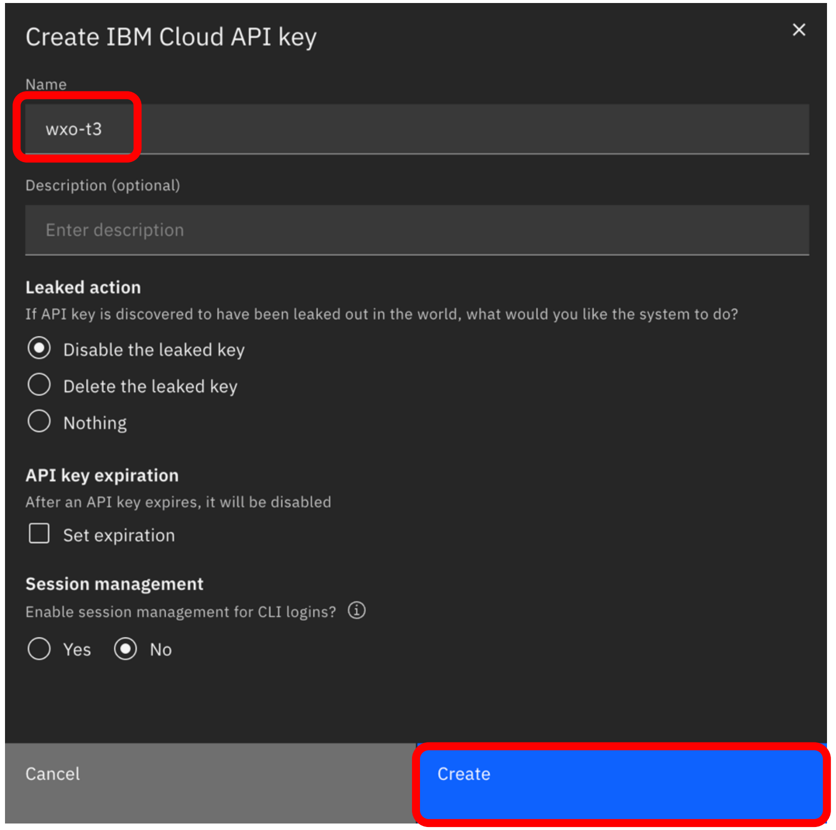


- Click on **Create** while “My IBM Cloud API keys” selected for the view

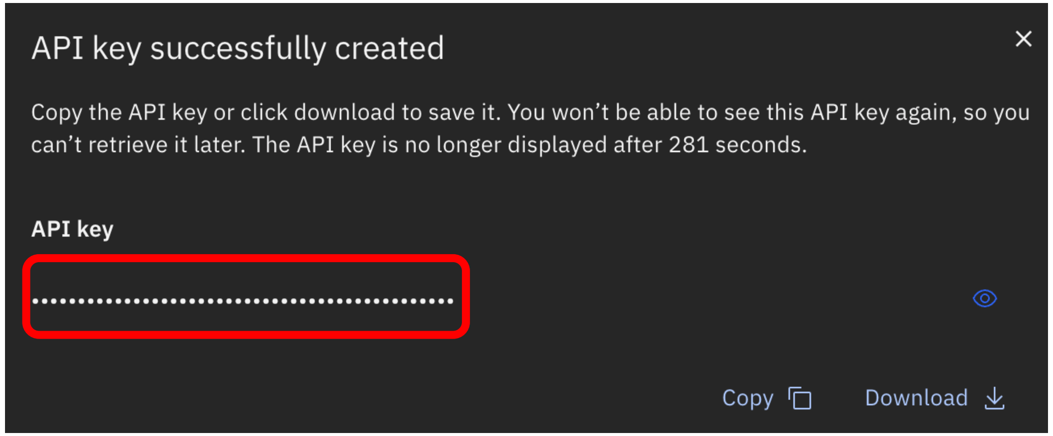


Option 2: Connect to wxO SaaS (6)

- Name the API as you wish, keep the default settings and click **Create**



- Copy the **API key value** and store it so that you can access it easily in the next steps



Option 2: Connect to wxO SaaS (4)

- As described in the generated instructions, you can run a simple command to add new environment to your ADK - for this open a terminal to your Bob IDE: **Terminal → New Terminal** (from the IBM Bob main menu bar)

- Run the command "**orchestrate env add -n <environment-name> -u <service-instance-url> --type ibm_iam**" replacing the <environment-name> with the *name you want to give to your environment (e.g. my-t3-wxo)* and the <service-instance-url> with *the URL you copied from your IBM Cloud wxO service front page* (look example below) and the new env will be created

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   IBM BOB FINDINGS   zsh

```
source /Users/jukkajuselius/bob-projects/environment-setup/.venv/bin/activate
jukkajuselius@Jukkas-iMac environment-setup % source /Users/jukkajuselius/bob-projects/environment-setup/.venv/bin/activate
(.venv) jukkajuselius@Jukkas-iMac environment-setup % orchestrate env add -n my-t3-wxo -u https://api.eu-de.watson-orchestrate.cloud.ibm.com/instances/da9308c8-056f-47b4-a954-5f9cd2ab2dc2  --type ibm_iam
[INFO] - Environment 'my-t3-wxo' has been created
(.venv) jukkajuselius@Jukkas-iMac environment-setup %
```

Option 2: Connect to wxO SaaS (5)

- You can now activate the environment running the command "**orchestrate env activate <environment-name>**", but for this you need your environment API key, copy it to your clipboard and paste it to terminal when asked

```
(.venv) jukkajuselius@Jukkas-iMac environment-setup % orchestrate env activate my-t3-wxo
Please enter WXO API key:
[INFO] - Environment 'my-t3-wxo' is now active
(.venv) jukkajuselius@Jukkas-iMac environment-setup %
```

- You are now good to go with the lab exercises!